

ZHENG Candi

 Hong Kong University of
Science and Technology

 Ph.D. in Mathematics

@ czhengac@connect.ust.hk

 <https://scraed.github.io/>

 Clear Water Bay, Kowloon,
Hong Kong

Education

2024.12–present	HKUST, Department of Mathematics, Postdoctoral Fellow
2019.02–2024.10	HKUST, Department of Mathematics, Ph.D. Candidate
2017.09–2019.02	HKUST, Department of Mechanical and Aerospace Engineering, Ph.D. Student
2016.09–2017.01	University of Edinburgh, Department of Physics, Visiting Student
2013.09–2017.06	SUSTech, Department of Physics, B.Sc.

Research Interests

- **Generative Diffusion Models:** Physically and mathematically motivated, more controllable diffusion generation methods. Focus on image inpainting, referencing, restoration, guidance, etc. Open-source projects: [LanPaint](#) (1K+ GitHub stars), [Characteristic Guidance Web UI](#) (84 stars).
- **Non-equilibrium Physics:** Particle kinetics and gas dynamics governed by Fokker-Planck and Boltzmann equations. Exploring AI applications in shock waves, light scattering, and constitutive relations.

Publications

Generative Diffusion Models

- **Candi Zheng**, Yuan Lan. “Rethinking the Diffusion Model from a Langevin Perspective.” The Blogpost Track at ICLR 2026.
- **Candi Zheng**[†], Yuan Lan[†], Yang Wang. “Lanpaint: Training-Free Diffusion Inpainting with Exact and Fast Conditional Inference.” TMLR, 2025.
- **Candi Zheng**[†], Yuan Lan[†]. “Characteristic Guidance: Non-linear Correction for DDPM at Large Guidance Scale.” Forty-first International Conference on Machine Learning (ICML). 2024.

Non-equilibrium Physics

- Florian Kogelbauer^{††}, **Candi Zheng**[†], Ilya Karlin. “Learning the Optimal Hydrodynamic Closure.” Submitted to PNAS, 2025.
- **Candi Zheng**^{*}, Yang Wang, and Shiyi Chen. “Phase Transition in Extended Thermodynamics Triggers Sub-shocks.” arXiv:2304.10742 (2023).
- **Candi Zheng**^{*}, Yang Wang, and Shiyi Chen. “Stabilizing the Maximal Entropy Moment Method for Rarefied Gas Dynamics at Single-Precision.” arXiv:2303.02898 (2023).

- › **Candi Zheng**^{*}, Yang Wang, and Shiyi Chen. “Data-driven constitutive relation reveals scaling law for hydrodynamic transport coefficients.” Physical Review E 107, no. 1 (2023): 015104.
- › Jin, Yuan-Jun, Rui Wang, Jin-Zhu Zhao, Yong-Ping Du, **Can-Di Zheng**, Li-Yong Gan, Jun-Feng Liu, Hu Xu, and S. Y. Tong. “The prediction of a family group of two-dimensional node-line semimetals.” Nanoscale 9, no. 35 (2017): 13112-13118.

^{*}Corresponding author [†]Equal contribution

Software Development

- › **LanPaint**: ComfyUI plugin for AI painting, 800+ GitHub stars. Enables precise inpainting with “deep thinking” without additional training; broadly compatible with various diffusion architectures and custom models.
- › **Characteristic Guidance Web UI**: Plugin for Stable Diffusion WebUI, providing image quality enhancement and high-controllability CFG scale generation.
- › **Moment Gauge**: JAX-based library for solving moment methods in gas dynamics.

Honors & Awards

- › Research Talent Hub, Innovation and Technology Fund, ITC Hong Kong (2025)
- › Top 10/2632, ATEC 2018 Developer AI Challenge (Financial Risk) Preliminary (2018)
- › Outstanding Graduation Design, Southern University of Science and Technology (2017)
- › Silver Medal, University Physics Competition (USA) (2015)
- › Third Prize, RoboMaster Central-South China Division (2015)
- › National Endeavor Scholarship (2014)
- › Outstanding Freshman Scholarship, SUSTech (2014)

Skills

- › **Languages**: English (Fluent), Chinese (Native)
- › **Mathematics & Physics**: Mathematical analysis/algebra, statistical physics, kinetic theory
- › **Programming**: Proficient in Python; experienced in ComfyUI plugin development; familiar with Fortran, Java; knowledge of CUDA C/MPI
- › **Machine Learning**: Proficient in PyTorch, JAX, and Scikit-Learn frameworks